A Cache in the Weights? Dynamic Evaluation and Temporal Locality in Small Language Models

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Abstract

Dynamic evaluation and test-time training let a language model keep learning from the text it reads at inference time. The cost is forgetting, as each update can overwrite earlier knowledge, and this is usually treated as a problem. However, the inputs a model receives at inference time (e.g. a book or a code base) often show some degree of temporal locality, i.e. recently seen content is likely to be seen again. In a small model, with little room in its weights, locality could make forgetting useful, so that together they could be exploited to obtain the behaviour of a cache memory. The weights would hold the recent context beyond the window, and new material would push out old material. We ask whether this cache behaviour arises on its own in a standard transformer that takes plain gradient steps as it reads. We test this with small Pythia models reading PG-19 books. We split the books into short chunks and score each one without the preceding text in context, so any help from earlier chunks must come through the weights. We then compare reading the chunks in order with reading them shuffled, which removes their locality. We find support for the cache hypothesis. The weights hold a memory of the current book beyond the attention window, stored mostly in the MLPs. This memory depends on the order of the input and fades as other books are read, halving within a few thousand tokens. As expected, it competes with general knowledge for the same weights. Knowledge of the same domain is largely preserved, while out-of-domain text is partly forgotten, and a stronger cache is hard to get without more of this forgetting. Decoupling the two could give a stronger cache and a better use of temporal locality, reducing the model size required for a given performance.

1 Introduction

Language models store in their parameters a vast amount of knowledge, most of which is irrelevant to any particular input they process. Small models are cheaper to run and can be deployed on memory-constrained devices, but their limited capacity restricts how much knowledge and context they can retain. Yet what a model reads at inference time has temporal locality, i.e. recently seen content is likely to be seen again: a book keeps its characters, places and turns of phrase for hundreds of thousands of tokens, and so do a long conversation or a code base, far longer than an attention window lasts. Computer memory hierarchies exploit this property with a small cache that holds recently used data and lets older data be pushed out, allowing a small memory to serve a workload as though it were a large one [12]. By analogy, a small model whose parameters cached its current context, instead of permanently holding everything it might ever need, could perform on that context comparably to a larger model.

A transformer already has such a cache in its attention window, but the window is short and everything before it is lost. Dynamic evaluation [24, 25, 34] and test-time training [13, 3, 38] let a language model keep learning from the text it reads at inference time, adapting its weights to the current input distribution and thus predicting what follows better than a frozen model. The cost is forgetting: each update can overwrite earlier knowledge, the catastrophic interference of sequential training [29], and this is usually treated as a problem. In a small model, with little room in its weights, locality and a forgetting that spares recent content could together give the behaviour of a cache memory: the weights would hold the recent context beyond the window, and new material would push out old material.

We ask whether this cache behaviour arises on its own in a standard transformer that takes plain gradient steps as it reads. If it does, three things should be true. The benefit should depend on *locality*: it should be larger when the input stays on the same material than when the same text arrives in a random order. New

material should push out old material, which we call *eviction*; this is the useful side of forgetting. And the cache should leave the knowledge the model already had largely intact; we call the loss of that general knowledge, the harmful side, simply *forgetting* from now on.

We test this with small Pythia models [6] reading PG-19 books [33]. We split the books into chunks of 256 tokens and score each one on its own, without the previous chunks in the context, so anything that helps from further back must come through the weights. After scoring a chunk, the model takes one update step on it. We compare reading the chunks in order with reading them shuffled, which removes their locality. Beyond the three predictions above, we study where in the network the memory lives, whether a knob sets its size, and whether it matters more in small models or can replace a larger one.

Our contributions are as follows.

- We measure the reach of the memory and its dependence on input locality, and compare it with a token-count cache (Sections 4.1 and 4.3).
- We locate where in the network the memory is stored (Section 4.2).
- We measure how fast other text evicts the memory, and how much general knowledge, inside and outside the domain, is forgotten (Sections 4.4 and 4.5).
- We study how the update rule and the model size affect the cache and its forgetting, and the trade-off between the two (Sections 4.6 and 4.7).
- We assess what dynamic evaluation is worth against a larger frozen model on long books (Section 4.8).

The long-term objective of this line of work is to determine whether such a cache can reduce the model size required for a given level of performance. The present study is a first step towards that objective, examining whether cache behaviour emerges in a standard transformer under dynamic evaluation and at what cost to general knowledge.

2 Related work

This work builds on the *temporal locality* that the inputs of language models exhibit to varying degrees, and on the possibility that a small model could exploit it by caching its current context in its weights through online learning. Temporal locality is a property of the input; exploiting it requires a recency bias in the model, which here resides in the weights rather than, as usually studied in language models, in the context window [42]. We review four related lines of research from this perspective.

Cache language models. Memory hierarchies exploit temporal locality with a small cache that retains recently used data, allowing a small memory to serve a workload almost as well as a large one [12]. Text exhibits the same property: a word that has occurred in a document is far more likely to recur than its corpus frequency predicts [9], and source code is similarly repetitive within a file or module [39]. Cache language models exploit this by interpolating the model's prediction with a distribution estimated from the recent history [26, 20]. Neural variants store recent hidden states instead of word counts [17, 21], and several transformer architectures extend the attention window with recurrent, compressed or external memories [11, 33, 41, 7]. In all of these the past is held in activations or in a separate store, and the parameters remain fixed. We instead restrict the context so that the past can only be held in the parameters, and ask whether they behave as a cache; a token-count cache serves as a baseline (Section 4.1).

Dynamic evaluation. Dynamic evaluation applies online learning to a trained model during evaluation: the parameters are updated by gradient descent on the portion of the sequence already observed, so that each token is predicted before it is trained on. It was first applied to recurrent language models [32, 18] and studied systematically by Krause et al. [24, 25], who motivate it by the recurrence of patterns within a sequence and favour recent history through a decay towards the trained weights. Rannen-Triki et al. [34] interpret it as a memory in the weights complementary to the context window, reset the weights at book boundaries, and show that under distribution shift a smaller model or context with online learning can offer a better trade-off between compute and loss than a larger static model. That dynamic evaluation benefits from temporal locality is thus the premise of this literature. Our contribution is to quantify it, comparing the same text in order and shuffled

and decomposing the gain by the distance and book of each token's previous occurrence (Sections 4.1 and 4.3). We use neither decay nor resets, so that eviction can be measured directly (Section 4.4), and we compare against model size rather than compute (Section 4.8).

Test-time training and fast weights. Test-time training adapts a model to each test instance through a self-supervised objective [36] and has inspired sequence layers and memory modules that learn as they read [37, 4]. Recent methods for transformer language models differ mainly in the parameters they update: the output projections of the MLP blocks [13], the query projections of attention [3], or the MLP weights of a model meta-trained for test-time adaptation [38]. They descend from fast weights, rapidly changing parameters that provide short-term memory on top of slow ones [19, 35, 2, 10]. These methods mostly target long contexts, fix by design which parameters adapt and how quickly the memory fades, and are evaluated by the reduction in loss. We study instead an unmodified pretrained model under plain gradient steps, and ask where the memory forms (Section 4.2), how the update rule and model size shape it (Section 4.6), and how long it persists, with a view to techniques that exploit temporal locality in small models.

Forgetting. When one set of parameters serves both short- and long-term memory, new learning interferes with existing knowledge. This catastrophic interference was first documented in connectionist networks trained sequentially [29, 14] and has motivated methods that protect the parameters most relevant to earlier tasks [23]. Large language models lose general capabilities during continual fine-tuning [28], models updated by dynamic evaluation degrade on older text [27], and knowledge acquired during pretraining decays as a power law of subsequent training steps [8]. Much of this knowledge appears to reside in the feed-forward layers [16, 30]. Forgetting may also be adaptive, reflecting the decreasing probability that an item will be needed again [1]; a cache relies on the same assumption, and its limited capacity can follow a changing context only by evicting what is no longer in use. We therefore measure both the eviction of a book's memory and the loss of general knowledge in the same runs (Sections 4.4, 4.5 and 4.7).

3 Setup

Every experiment in this paper has the same structure. A model reads an *input stream*: a single, ordered sequence of text that it reads once, from beginning to end, in consecutive chunks of 256 tokens, as a language model reads text during inference. Attention covers only the current chunk, so whatever the model can use from earlier parts of the stream must be carried in its weights. Under dynamic evaluation the model updates those weights as it reads; the frozen pretrained model reading the same stream without updates is the reference. The experiments differ in the streams they use, which vary in length and in how often their content recurs, and in what is measured on the updated weights along the way. We first describe the streams and why each is needed, then how the learner reads them and what we measure.

3.1 Models

We use Pythia-14M, 31M and 70M [6], with 14.07M, 30.49M and 70.43M parameters respectively; all three have 6 layers and share the same tokenizer, and all runs use float32 on CPU with seed 0.

3.2 Input streams

All streams are built from books of PG-19 [33]. Each book is tokenized on its own and cut into non-overlapping chunks of 256 tokens, the remainder dropped, so that no chunk spans two books. A stream is a set of such chunks placed in a fixed reading order. It is read without interruption: the learner starts from the pretrained weights at the beginning of the stream, and its weights are not reset between books, so a book is read with whatever the weights retain from the books before it.

Different questions call for different streams, and we use four kinds, summarized in Table 1.

Main streams: the same text in different orders. The central question is whether the learner exploits the temporal locality of text, that is, whether its gain depends on having recently read related material. Answering it requires streams that hold the same text in different orders, so that any difference between them is due to the order alone. The *main set* consists of 25 test books, each cut to its first 78 chunks (19,968 tokens), 1,950

stream	books	chunks	order	used for
main ($K = 1, 5, 25$, shuffled)	25 test, 20k tokens each	1,950	four orders	locality, most results
long	5 test, 100k tokens each	1,950	book after book	locality on longer books
very long	4 test, 200k tokens each	3,124	book after book	larger frozen model, forgetting
val-pick	10 validation, 20k tokens each	780	book after book	learning rates only
eviction (one per book)	half a test book, then others	varies	book first	eviction

Table 1: The input streams. Every stream is read once, chunk by chunk, starting from the pretrained weights.

chunks in total. Its chunks are arranged in four orders of decreasing locality: book after book ($K = 1$), as a book is normally read; groups of 5 books, interleaved chunk by chunk within each group ($K = 5$); all 25 books interleaved chunk by chunk ($K = 25$); and a single random permutation of all chunks (shuffled), in which no book is read in order. Cutting the books short keeps the stream at about 500k tokens while leaving 25 books to compute standard errors over, and each book is still long enough to measure what the learner retains from up to about 10k tokens back within it. Most results use the main streams, read in order and shuffled; the interleaved orders appear in Section 4.3.

Long streams: reading further into a book. If the weights act as a cache of the current context, the gain should grow the longer the input stays on the same material. The main streams can test this only in one direction: their most local order, book after book, still changes book every 20k tokens, and the other orders lower locality further by interleaving books. The *long* stream tests the other direction. It holds 5 test books of 100k tokens (390 chunks each), read book after book, so that the input stays on the same book five times longer and most tokens are read with tens of thousands of tokens of the same book already behind them; it forms the most local step of the comparison in Section 4.3. The *very long* stream, 4 test books of 200k tokens (781 chunks each), also read book after book, compares dynamic evaluation with a larger frozen model over a whole long book (Section 4.8) and tracks forgetting over the longest stream (Section 4.5). Both hold fewer books than the main set, so their standard errors are wider. All 5 long books and 2 of the 4 very long books also appear in the main set, cut there to their first 78 chunks.

The val-pick stream: tuning without the test books. The learning rate has a different scale under each update rule, and the step a learner tolerates also depends on which parameters are updated and on model size, so each combination of rule, subset and size requires its own rate. To avoid selecting rates on the evaluation data, all rates are selected on a separate stream, *val-pick*: 10 PG-19 validation books of 78 chunks each, read book after book, matching the main set in book length and reading order. No reported result uses val-pick.

Eviction streams: one book, then other text. A cache must also let go of old content as new content arrives. The main streams cannot measure this directly, because a book is never read again after the stream moves on, and the amount of other text read in between is not controlled. The eviction experiment therefore gives each of 20 main-set books its own short stream: the first half of the book, followed by a controlled amount of other books. At the end of each stream, a frozen copy of the learner is tested on the unread second half of the book, which shows how much of the book's memory survives that amount of other text (Section 3.4).

Held-out text. Updating the weights may also erase general knowledge the model had before reading the stream. The streams cannot measure this, since every chunk in them is trained on after it is scored. We therefore use two held-out sets that are never part of any stream and are never trained on: 76 chunks of the detokenized WikiText-103 test split [31], outside the domain, and 6 other PG-19 validation books of 15 chunks each, inside it. They are only scored, by frozen copies of the learner (Section 3.4), so that losses on them measure knowledge lost, both of text unlike the books and of books like those being read.

A gain on the test books could also come from text the model saw during pretraining rather than from the stream. Pythia was trained on the Pile [15], whose PG-19 component has 28,602 documents [5], the size of PG-19's training split. The test books are therefore very likely absent from Pythia's training data, although the Pile's authors do not state this explicitly and we did not check for other copies of the books.

3.3 Dynamic evaluation

The learner reads a stream one chunk at a time, in two steps per chunk. It first scores the chunk on its own, without earlier chunks in the context, so any benefit from earlier parts of the stream must come through the weights. It then takes one update step on the chunk's mean loss and moves on to the next chunk. Every token is therefore predicted before it is trained on. We call the updated model the *learner*, and compare it with the frozen pretrained model reading the same stream without updates.

We compare three update rules. *Plain SGD* uses no momentum or weight decay; *Adam* [22] starts each stream with a fresh optimizer state; and *normalized SGD* moves every tensor by the same relative amount per step, $W \leftarrow W - \eta \, \|W\| \, g/\|g\|$. All three update all parameters. In addition, some plain-SGD runs on Pythia-14M update only part of the model: the output embedding, everything but the output embedding, the MLPs of all layers, the attention of all layers, or the MLP of a single layer. Each combination of update rule, parameter subset and model size has its own learning rate, chosen as the value with the lowest mean loss under dynamic evaluation on the val-pick stream, from a grid of 1 and 3 times powers of ten. The grid is extended until both neighbours of the best value have been tried, so every selected rate lies inside its grid.

3.4 Snapshots

To see what the weights hold at a given point of a stream, we take *snapshots*: the learner's weights are copied and the copy, kept frozen, scores text outside the stream, while the learner continues reading unaffected. For forgetting, snapshots are taken every 40 chunks and scored on the held-out sets. For eviction (Section 4.4), each of 20 main-set books A is split into a first half A1 (40 chunks) and a second half A2 (38 chunks), and gets a stream of its own, read from the pretrained weights: A1 followed by I tokens of PG-19 validation books. At the end of this stream a snapshot scores A2 and the same span C2 of a control book C that the learner never read.

3.5 Metrics

The *gain* measures how much dynamic evaluation helps: the frozen loss minus the loss under dynamic evaluation on the same tokens of the stream, in nats/token, always against the model's own frozen loss. To see where the gain comes from, *distance bins* group tokens by how far back in the stream the same token id last occurred: in the same chunk (visible to attention), in an earlier chunk less than 1k, 1k to 10k, 10k to 100k or at least 100k tokens back, or never. The 1000 most frequent ids are left out of the earlier-chunk bins, which are further split by whether the previous occurrence was in the same book or an earlier one. *Locality* is the part of the gain that requires the reading order: the gain on a stream in reading order minus the gain on the same chunks shuffled. In the eviction experiment, the *memory of A* is what a snapshot still knows about book A beyond general adaptation: its gain on A2 minus its gain on C2. *Forgetting* is the loss of general knowledge: a snapshot's loss on a held-out set minus that of the pretrained model, averaged over the snapshots in the second half of the stream, so that positive values mean knowledge lost. Finally, the *exchange rate* relates the two: a run's gain on the stream divided by its WikiText forgetting.

Means are taken over tokens. Standard errors (SE) are computed over books, from per-book means or, when two runs read the same chunks, from paired per-book differences; books with fewer than 50 tokens in a bin are left out. For forgetting, SE are computed over the WikiText chunks or the 6 held-out books. A † marks a value under 2 SE, which we do not treat as a difference.

4 Results

4.1 A memory beyond the window, and it is local

If the updated weights hold a memory of the text, tokens seen recently in an earlier chunk should be predicted better, and the benefit should depend on the order of the chunks. We bin tokens by the distance to their previous occurrence and compare reading in order with reading the same chunks shuffled.

With plain SGD at lr 10^{-4}, Pythia-14M gains $+0.100 \pm 0.006$ nats/token over its frozen loss of 4.393 (Table 2). The gain falls with distance: $+0.307 \pm 0.011$ for rare tokens last seen less than 1k tokens back, $+0.144 \pm 0.008$ between 1k and 10k, and $+0.009 \pm 0.006^{\dagger}$ between 10k and 100k. Tokens never seen before lose -0.059 ± 0.013, and tokens repeated within their own chunk lose -0.111 ± 0.014. Within a book the gain grows from $+0.013 \pm 0.008^{\dagger}$

bin (previous occurrence)	tokens	frozen loss	in order	shuffled	locality	unigram cache
all tokens	497,250	4.393	$+0.100_{\pm.006}$	$+0.062_{\pm.003}$	$+0.038_{\pm.004}$	$+0.018_{\pm.004}$
same chunk	9,886	5.006	$-0.111_{\pm.014}$	$-0.133_{\pm.010}$	$+0.022^{\dagger}_{\pm.013}$	$+0.096_{\pm.017}$
earlier, $<$ 1k	17,036	7.335	$+0.307_{\pm.011}$	$+0.062_{\pm.007}$	$+0.245_{\pm.009}$	$+0.461_{\pm.024}$
earlier, 1k-10k	33,973	6.997	$+0.144_{\pm.008}$	$+0.035_{\pm.005}$	$+0.109_{\pm.006}$	$+0.059_{\pm.007}$
earlier, 10k-100k	44,109	7.113	$+0.009^{\dagger}_{\pm.006}$	$+0.026_{\pm.005}$	$-0.017_{\pm.006}$	$-0.051_{\pm.000}$
earlier, $\geq$ 100k	13,125	7.824	$-0.039_{\pm.009}$	$-0.017_{\pm.008}$	$-0.023_{\pm.008}$	$-0.051_{\pm.000}$
never seen	20,976	8.190	$-0.059_{\pm.013}$	$-0.055_{\pm.009}$	$-0.004^{\dagger}_{\pm.010}$	$-0.051_{\pm.000}$

Table 2: Gain over frozen Pythia-14M by distance to the token's previous occurrence (nats/token, ± SE over 25 books). Dynamic evaluation: plain SGD on all parameters, lr 10^{-4}, in reading order ($K = 1$) and with the chunks shuffled; locality = in order − shuffled. Unigram cache: frozen model mixed with a decaying token-count cache; where the cache has no count for a token, mixing it in costs exactly $-\log 0.95 = 0.051$ nats, so those bins have a constant loss and an SE of 0. †: under 2 SE.

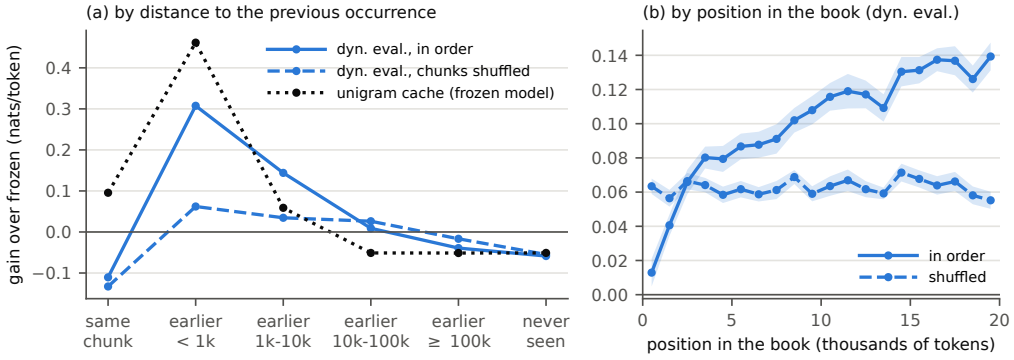


Figure 1: Gain over frozen Pythia-14M with plain SGD (lr 10^{-4}). (a) By distance to the token's previous occurrence, for dynamic evaluation in order and shuffled and for the unigram cache. (b) By position in the book, mean over the 24 books that follow a book switch, ±1 SE over books.

in its first 1k tokens to $+0.139 \pm 0.008$ in its last 1k, while the shuffled learner's gain stays between +0.055 and +0.071 (Figure 1b).

Shuffling keeps 62% of the overall gain, but only 20% less than 1k tokens back and 24% between 1k and 10k (locality $+0.245 \pm 0.009$ and $+0.109 \pm 0.006$). Most of the overall gain is therefore adaptation to PG-19 text in general, and most of the long-range gain needs the order.

The current book. A token may be predicted better because it was seen recently, or because it belongs to the book being read. Splitting the earlier-chunk bins by book separates the two (Table 3). Between 1k and 10k tokens back, the locality is $+0.120 \pm 0.007$ for tokens last seen in the same book and $+0.035 \pm 0.009$ for tokens last seen in an earlier book. Between 10k and 100k tokens back, same-book tokens keep $+0.036 \pm 0.010$ of locality and earlier-book tokens have none (-0.024 ± 0.006). The memory is tied mostly to the current book, although distance still matters within it.

More than a count cache. A cache of recent token counts mixed into the frozen model ($p = 0.95\, p_{\text{model}} + 0.05\, p_{\text{cache}}$, counts halving every 1k tokens) is the simplest way to use the recent past. It is the better copier less than 1k tokens back ($+0.461 \pm 0.024$ against +0.307), but from 1k tokens on dynamic evaluation gains more (+0.144 against $+0.059 \pm 0.007$ between 1k and 10k; paired difference $+0.085 \pm 0.008$), and $+0.082 \pm 0.004$ more overall (Table 2).

4.2 Where the memory lives

If the memory sits in particular weights, updating only those should keep it. We update one part of Pythia-14M at a time, each part at its own learning rate (Table 4).

bin	previous occurrence	tokens	in order	shuffled	locality	unigram cache
earlier, $<$ 1k	same book	16,967	$+0.308_{\pm.011}$	$+0.062_{\pm.007}$	$+0.246_{\pm.009}$	$+0.462_{\pm.024}$
earlier, 1k-10k	same book	29,577	$+0.154_{\pm.008}$	$+0.034_{\pm.005}$	$+0.120_{\pm.007}$	$+0.070_{\pm.009}$
earlier, 1k-10k	earlier book	4,396	$+0.076_{\pm.012}$	$+0.040_{\pm.009}$	$+0.035_{\pm.009}$	$-0.016_{\pm.004}$
earlier, 10k-100k	same book	4,798	$+0.060_{\pm.011}$	$+0.024_{\pm.007}$	$+0.036_{\pm.010}$	$-0.050_{\pm.000}$
earlier, 10k-100k	earlier book	39,311	$+0.003^{\dagger}_{\pm.006}$	$+0.026_{\pm.005}$	$-0.024_{\pm.006}$	$-0.051_{\pm.000}$

Table 3: Gain over frozen Pythia-14M by whether the token's previous occurrence is in the same book or an earlier one (± SE over books); same runs as Table 2. The earlier-book part of the $<$ 1k bin (69 tokens) is omitted. †: under 2 SE.

parameters updated	parameters	lr	all tokens	earlier, $<$ 1k	1k-10k, same book	never seen
all parameters	14,067,712	10^{-4}	$+0.100_{\pm.006}$	$+0.307_{\pm.011}$	$+0.154_{\pm.008}$	$-0.059_{\pm.013}$
output embedding only	6,438,912	3×10^{-3}	$+0.072_{\pm.005}$	$+0.252_{\pm.017}$	$+0.072_{\pm.006}$	$-0.143_{\pm.012}$
all but output embedding	7,628,800	10^{-4}	$+0.094_{\pm.005}$	$+0.294_{\pm.011}$	$+0.151_{\pm.009}$	$-0.058_{\pm.013}$
MLPs only	790,272	3×10^{-4}	$+0.107_{\pm.007}$	$+0.415_{\pm.016}$	$+0.175_{\pm.011}$	$-0.077_{\pm.014}$
attention only	396,288	3×10^{-4}	$+0.066_{\pm.005}$	$+0.259_{\pm.011}$	$+0.110_{\pm.009}$	$-0.074_{\pm.014}$
MLP of layer 1 only	131,712	10^{-3}	$+0.053_{\pm.004}$	$+0.236_{\pm.012}$	$+0.113_{\pm.007}$	$-0.055_{\pm.010}$
MLP of layer 2 only	131,712	10^{-3}	$+0.069_{\pm.004}$	$+0.242_{\pm.011}$	$+0.131_{\pm.008}$	$-0.051_{\pm.010}$
MLP of layer 3 only	131,712	3×10^{-4}	$+0.055_{\pm.004}$	$+0.204_{\pm.009}$	$+0.093_{\pm.007}$	$-0.056_{\pm.011}$
MLP of layer 4 only	131,712	10^{-3}	$+0.063_{\pm.004}$	$+0.187_{\pm.010}$	$+0.101_{\pm.007}$	$-0.063_{\pm.010}$
MLP of layer 5 only	131,712	3×10^{-3}	$+0.073_{\pm.006}$	$+0.285_{\pm.014}$	$+0.121_{\pm.009}$	$-0.090_{\pm.014}$
MLP of layer 6 only	131,712	10^{-3}	$+0.048_{\pm.004}$	$+0.111_{\pm.008}$	$+0.053_{\pm.006}$	$-0.084_{\pm.012}$

Table 4: Gain over frozen Pythia-14M when only some parameters are updated with plain SGD, in reading order ($K = 1$), each subset at its own val-pick learning rate (± SE over books). Columns are distance bins; "1k-10k, same book" counts tokens whose previous occurrence is in the same book. †: under 2 SE.

Freezing the output embedding costs little (-0.006 ± 0.001 overall against all parameters). The output embedding alone is a short-range cache: $+0.252 \pm 0.017$ less than 1k tokens back but $+0.072 \pm 0.006$ between 1k and 10k (same book), with the largest loss on never-seen tokens (-0.143 ± 0.012). Attention alone keeps less of the long-range gain ($+0.110 \pm 0.009$ between 1k and 10k, same book). Updating only the MLPs (lr 3×10^{-4}) gains more than updating all parameters (lr 10^{-4}): $+0.415 \pm 0.016$ against $+0.307$ less than 1k tokens back (paired difference $+0.108 \pm 0.008$) and $+0.107 \pm 0.007$ against $+0.100$ overall ($+0.007 \pm 0.002$).

One layer's MLP alone keeps between 36% and 93% of the full learner's gain less than 1k tokens back, and between 35% and 85% between 1k and 10k (same book); every single layer is below the full learner there (Figure 2a). The strongest is layer 5 ($+0.285 \pm 0.014$, paired difference with all parameters -0.023 ± 0.009), and the last layer is the weakest ($+0.111 \pm 0.008$).

MLPs only against all parameters. A larger gain could come from a stronger cache or from better general adaptation, and a stronger cache might cost more general knowledge. We read the shuffled stream with both learners and probe both with snapshots (Table 5). Updating only the MLPs gives more locality: $+0.334 \pm 0.011$ against $+0.245 \pm 0.009$ less than 1k tokens back (paired difference $+0.088 \pm 0.005$) and $+0.145 \pm 0.009$ against $+0.120 \pm 0.007$ between 1k and 10k, same book ($+0.025 \pm 0.003$). Shuffled, it gains slightly less overall (-0.003 ± 0.001). It also forgets more: WikiText forgetting is $+0.235 \pm 0.007$ against $+0.196 \pm 0.006$ (paired $+0.039 \pm 0.003$), and the held-out books lose $+0.022 \pm 0.002$ more (paired), although neither run's own value on them is beyond 2 SE. Its exchange rate is lower: 0.45 against 0.51 in order, 0.41 against 0.55 shuffled.

The two learners use different learning rates, each its own val-pick (val-pick loss 4.2506 for the MLPs at 3×10^{-4} and 4.2565 for all parameters at 10^{-4}). All parameters at 3×10^{-4} is already worse on val-pick (4.2732). A possible reason, not tested: with plain SGD at 14M, attention takes 43% of the squared update norm and the MLPs 50% (Section 4.6), which may limit the step the full learner can use.

4.3 Locality as a variable

A cache should help more the longer the input stays on the same material. We read the same chunks with more and more books interleaved, and the long books in order (Figure 2b).

With plain SGD on all parameters at lr 10^{-4}, the gain falls at every step: $+0.176 \pm 0.023$ on the long books, $+0.100 \pm 0.006$ with $K = 1$, $+0.074 \pm 0.004$ with $K = 5$, $+0.065 \pm 0.003$ with $K = 25$ and $+0.062 \pm 0.003$ shuffled.

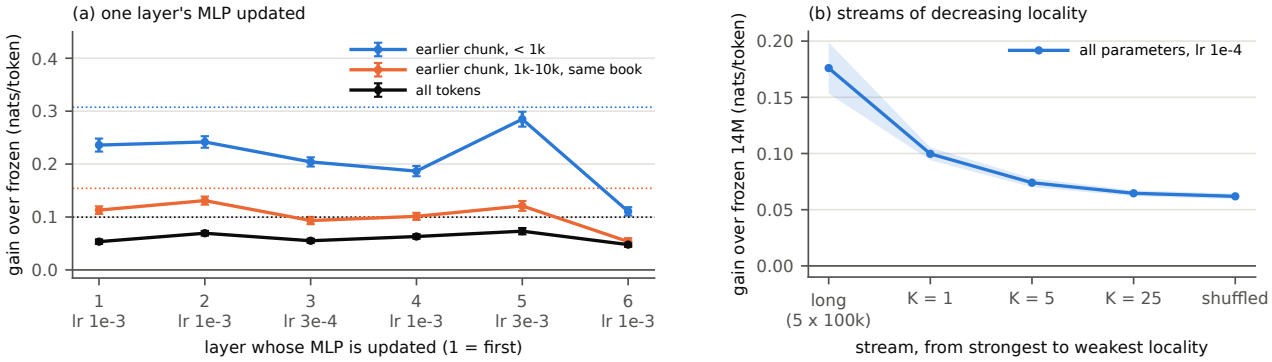


Figure 2: Gain over frozen Pythia-14M with plain SGD, ±1 SE over books. (a) Only one layer's MLP updated, each at its own learning rate (below the layer number); dotted lines: all parameters at lr 10^{-4}. (b) All parameters at lr 10^{-4}, on streams of decreasing locality: 5 long books of 100k tokens, then the main set with books in order ($K = 1$), 5 or 25 books interleaved, and shuffled.

quantity	bin / stream	MLPs only, lr 3×10^{-4}	all, lr 10^{-4}	MLPs only − all
gain shuffled	all tokens	$+0.059_{\pm.003}$	$+0.062_{\pm.003}$	$-0.003_{\pm.001}$
	earlier, < 1k	$+0.081_{\pm.011}$	$+0.062_{\pm.007}$	$+0.019_{\pm.005}$
	1k-10k, same book	$+0.030_{\pm.006}$	$+0.034_{\pm.005}$	$-0.004^{\dagger}_{\pm.002}$
	never seen	$-0.071_{\pm.010}$	$-0.055_{\pm.009}$	$-0.016_{\pm.003}$
locality	all tokens	$+0.048_{\pm.004}$	$+0.038_{\pm.004}$	$+0.010_{\pm.001}$
	earlier, < 1k	$+0.334_{\pm.011}$	$+0.245_{\pm.009}$	$+0.088_{\pm.005}$
	1k-10k, same book	$+0.145_{\pm.009}$	$+0.120_{\pm.007}$	$+0.025_{\pm.003}$
	never seen	$-0.006^{\dagger}_{\pm.011}$	$-0.004^{\dagger}_{\pm.010}$	$-0.002^{\dagger}_{\pm.004}$
WikiText forgetting	$K = 1$	$+0.235_{\pm.007}$	$+0.196_{\pm.006}$	$+0.039_{\pm.003}$
	shuffled	$+0.144_{\pm.005}$	$+0.112_{\pm.004}$	$+0.031_{\pm.002}$
held-out books forgetting	$K = 1$	$+0.016^{\dagger}_{\pm.009}$	$-0.006^{\dagger}_{\pm.010}$	$+0.022_{\pm.002}$
	shuffled	$-0.034_{\pm.007}$	$-0.045_{\pm.008}$	$+0.011_{\pm.001}$
exchange rate	$K = 1$	0.45	0.51	−0.06
	shuffled	0.41	0.55	−0.14

Table 5: Pythia-14M with plain SGD updating only the MLPs or all parameters, each at its own val-pick learning rate. Gain shuffled and locality (in order − shuffled) by distance bin, ± SE over books; forgetting (snapshot − pretrained loss, mean over the second half of the stream) on detokenized WikiText and the 6 held-out books, ± SE over held-out chunks or books; exchange rate = stream gain / WikiText forgetting. Last column: paired difference. The in-order gains are in Table 4. †: under 2 SE.

On the main set the paired drops are -0.026 ± 0.002 from $K = 1$ to $K = 5$, -0.009 ± 0.002 from $K = 5$ to $K = 25$ and -0.003 ± 0.001 from $K = 25$ to shuffled; interleaving only 5 books removes most of the in-order advantage. The long books are other books than the main set, so the step from them to $K = 1$ is not paired.

4.4 Eviction

If dynamic evaluation acts as a cache, reading other text should push a book out of memory. We let the model read the first half of a book, then other books, and check with a frozen snapshot how well it still predicts the book's second half, compared with a book it never read (Figure 3).

With plain SGD on all parameters at lr 10^{-4}, the memory of A is $+0.097 \pm 0.013$ right after A1. It halves after about 6.1k tokens of other books and then declines slowly: $+0.044 \pm 0.007$ after 10k tokens, $+0.031 \pm 0.006$ after 100k and $+0.016 \pm 0.008$ after 300k. The snapshots never gain on the control book (-0.030 ± 0.007 right after A1, between -0.032 and -0.011 later), so the decline is the loss of A rather than general adaptation catching up. The memory is mostly about tokens that occurred in A1: rare tokens of A2 whose id occurs in A1 carry $+0.223 \pm 0.029$ after A1, and other rare tokens $+0.056 \pm 0.011$; the former halve after about 2.9k tokens. A power law in I fits the memory better than an exponential with a floor (χ^2 2.0 against 8.7, with 4 degrees of freedom each), but the 7 points share the same 20 books, so this only ranks the two shapes.

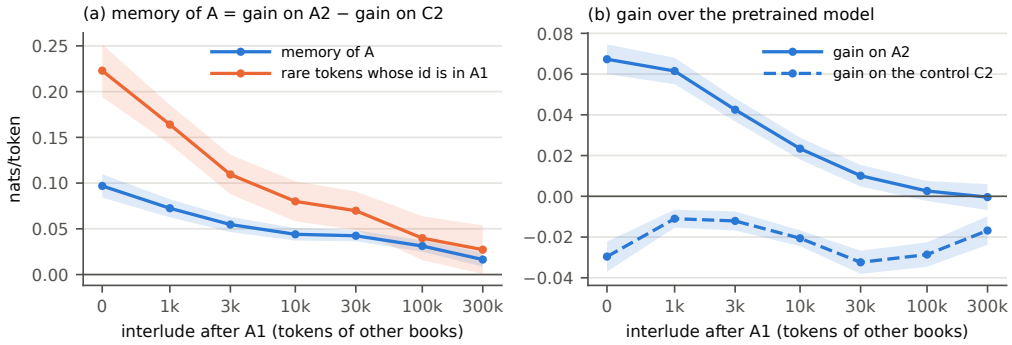


Figure 3: Eviction probed with frozen snapshots (Pythia-14M, plain SGD, all parameters, lr 10^{-4}; mean of 20 books, bands ±1 SE over books), against the number of tokens of other books read after A1. (a) Memory of A (gain on A2 − gain on the never-read control C2, same snapshot), and the same for rare tokens whose id occurs in A1. (b) Gain over the pretrained model on A2 and on C2.

stream	stream gain	WikiText forgetting	held-out books forgetting	exchange rate
$K = 1$	$+0.100_{\pm.006}$	$+0.196_{\pm.006}$	$-0.006^{\dagger}_{\pm.010}$	0.51
shuffled	$+0.062_{\pm.003}$	$+0.112_{\pm.004}$	$-0.045_{\pm.008}$	0.55
very long (4 × 200k)	$+0.170_{\pm.011}$	$+0.268_{\pm.008}$	$+0.052_{\pm.013}$	0.63

Table 6: Forgetting of held-out text by Pythia-14M with plain SGD on all parameters (lr 10^{-4}): snapshot loss − pretrained loss, mean over the snapshots in the second half of the stream, ± SE over the WikiText chunks or the 6 held-out books; positive means knowledge lost. Stream gain: over the frozen loss on the stream, ± SE over books. Exchange rate = stream gain / WikiText forgetting. †: under 2 SE.

4.5 Forgetting

A cache should leave the rest of memory intact. We score frozen snapshots of the learner on text it never reads: WikiText, outside the domain, and unseen PG-19 books, inside it (Table 6).

With plain SGD at lr 10^{-4}, Pythia-14M reading the main set in order forgets $+0.196 \pm 0.006$ nats/token on WikiText (pretrained loss 4.633), about 2.0 times its gain on the stream (exchange rate 0.51). On the held-out PG-19 books it loses nothing ($-0.006 \pm 0.010^{\dagger}$). The shuffled learner forgets less on WikiText ($+0.112 \pm 0.004$) and improves on the unseen books (-0.045 ± 0.008): the part of the gain that survives shuffling is adaptation to PG-19 text, and it carries over to unseen books. Reading in order forgets more than shuffled on both sets (shuffled − in order: -0.084 ± 0.003 on WikiText, -0.040 ± 0.002 on the books), at about the same exchange rate (0.51 and 0.55). On the very long set (800k tokens), WikiText forgetting averages +0.143 over the snapshots taken in the first book and +0.273, +0.269 and +0.267 in the next three: it levels off after the first book. There the held-out books also lose a little ($+0.052 \pm 0.013$).

4.6 Update rules and model size

With plain SGD, a learning rate small enough for some weights may be too small for others, and the balance changes with model size. We log the size of each update and compare three update rules at three sizes, each with its own learning rate (Table 7, Figure 4).

Over the first 40 chunks, plain SGD at each size's pick moves the MLP weights by 1.2×10^{-4} of their norm per step at 14M, 9.9×10^{-5} at 31M and 1.1×10^{-6} at 70M, about 110 times less at 70M than at 14M; at 70M attention takes 78% of the squared update norm and the MLPs 4%. With plain SGD, Pythia-70M gains $+0.025 \pm 0.002$ and has almost no locality ($+0.016 \pm 0.003$ less than 1k tokens back). Normalized SGD more than doubles 70M's gain ($+0.057 \pm 0.004$; paired difference $+0.033 \pm 0.002$) and gives it locality ($+0.084 \pm 0.006$), still far less than at 14M and 31M under the same rule ($+0.372 \pm 0.011$ and $+0.418 \pm 0.013$), which are the largest of any rule. Adam has no cache at any size: its locality less than 1k tokens back is -0.052 ± 0.009 at 14M, -0.070 ± 0.011 at 31M and -0.140 ± 0.010 at 70M.

From 14M to 31M, plain SGD's locality does not change beyond noise (31M − 14M: $+0.017 \pm 0.014^{\dagger}$ less than 1k tokens back, $+0.002 \pm 0.006^{\dagger}$ overall; SE unpaired across models); it falls at 70M under every rule. The

size	rule	lr	gain, $K = 1$	locality, < 1k	locality, all	WikiText forgetting	exch. rate, $K = 1$	shuffled
14M	plain SGD	10^{-4}	$+0.100_{\pm.006}$	$+0.245_{\pm.009}$	$+0.038_{\pm.004}$	$+0.196_{\pm.006}$	0.51	0.55
	Adam	3×10^{-6}	$+0.086_{\pm.005}$	$-0.052_{\pm.009}$	$-0.004^{\dagger}_{\pm.004}$	$+0.141_{\pm.005}$	0.61	0.91
	normalized SGD	10^{-4}	$+0.113_{\pm.008}$	$+0.372_{\pm.011}$	$+0.056_{\pm.006}$	$+0.316_{\pm.008}$	0.36	0.29
31M	plain SGD	3×10^{-4}	$+0.085_{\pm.005}$	$+0.262_{\pm.010}$	$+0.040_{\pm.004}$	$+0.228_{\pm.005}$	0.37	0.31
	Adam	3×10^{-6}	$+0.070_{\pm.004}$	$-0.070_{\pm.011}$	$-0.007_{\pm.003}$	$+0.137_{\pm.004}$	0.51	0.75
	normalized SGD	10^{-4}	$+0.092_{\pm.007}$	$+0.418_{\pm.013}$	$+0.060_{\pm.005}$	$+0.348_{\pm.006}$	0.26	0.13
70M	plain SGD	10^{-5}	$+0.025_{\pm.002}$	$+0.016_{\pm.003}$	$+0.004_{\pm.002}$	$+0.010_{\pm.002}$	2.57	2.67
	Adam	3×10^{-7}	$+0.029_{\pm.002}$	$-0.140_{\pm.010}$	$-0.015_{\pm.002}$	$+0.060_{\pm.003}$	0.47	0.68
	normalized SGD	3×10^{-6}	$+0.057_{\pm.004}$	$+0.084_{\pm.006}$	$+0.018_{\pm.003}$	$+0.039_{\pm.002}$	1.48	1.52

Table 7: Three update rules at three sizes, all parameters, each at its val-pick learning rate: gain with $K = 1$ over each model's frozen loss and locality (in order − shuffled, paired), ± SE over books; WikiText forgetting ($K = 1$, second half of the stream, ± SE over chunks); exchange rate (stream gain / WikiText forgetting) in order and shuffled. †: under 2 SE.

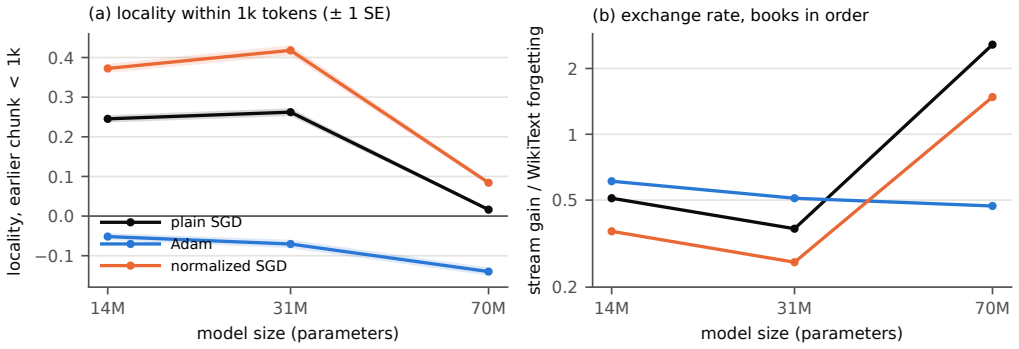


Figure 4: (a) Locality less than 1k tokens back against model size for three update rules, each at its own learning rate, ±1 SE over books. (b) Exchange rate (stream gain / WikiText forgetting, $K = 1$; log scale).

smaller model does not pay more general knowledge per nat of gain: the exchange rate at 14M is 0.51, 0.61 and 0.36 (plain SGD, Adam, normalized SGD), against 0.37, 0.51 and 0.26 at 31M. At 70M it is 2.57, 0.47 and 1.48, with much smaller gains.

4.7 A trade-off between cache and general knowledge

If the cache and general knowledge share the same weights, filling one should erode the other. We place every run with snapshots on two axes: how much cache it has (locality less than 1k tokens back) and how much WikiText it forgets (Figure 5).

Within 14M the order is the same on both axes: Adam (−0.052, +0.141), plain SGD on all parameters (+0.245, +0.196), plain SGD on the MLPs only (+0.334, +0.235) and normalized SGD (+0.372, +0.316). Within 31M the same holds for Adam, plain SGD and normalized SGD. At 70M plain SGD and normalized SGD follow it, but Adam forgets more than both ($+0.060 \pm 0.003$) with negative locality. Across sizes the points do not lie on one curve: at 70M every rule forgets far less for a given locality. With ten points, one seed and one model family, this is a consistent ordering within the 14M and 31M settings we tried, not a law, and it does not hold for Adam at 70M.

4.8 Long books and model size

A cache is worth more the longer the material stays the same, and a small model might use it to approach a larger one. We let Pythia-14M and Pythia-31M read books of 200k tokens with dynamic evaluation and compare them with frozen 14M, 31M and 70M (Table 8, Figure 6).

With plain SGD at lr 10^{-4}, Pythia-14M gains $+0.170 \pm 0.011$ over its frozen loss and closes 49% of the gap to a frozen Pythia-31M. The share grows from 7% in the first 10k tokens of a book to 79% between 160k and 170k tokens, and is 56% in the last 10k; frozen 31M stays ahead overall ($+0.180 \pm 0.019$). Pythia-31M with dynamic

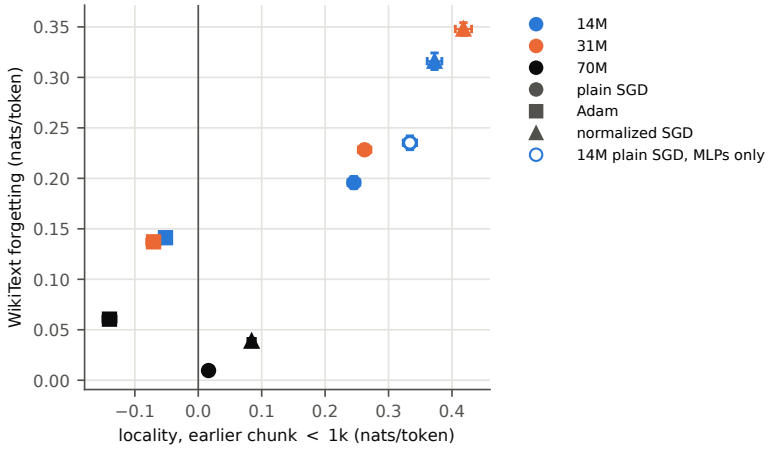


Figure 5: Locality less than 1k tokens back against WikiText forgetting ($K = 1$, second half of the stream) for every run with snapshots: three update rules (marker) at three sizes (colour) on all parameters, and plain SGD on the MLPs only at 14M (open circle); each at its own learning rate. Bars: ±1 SE.

run	loss	gain over frozen 14M	share of 14M-31M gap
frozen 14M	4.419		
14M dyn. eval., lr 10^{-4}	4.249	$+0.170_{\pm.011}$	49%
frozen 31M	4.069	$+0.351_{\pm.010}$	100%
31M dyn. eval., lr 3×10^{-4}	3.901	$+0.518_{\pm.013}$	148%
frozen 70M	3.846	$+0.573_{\pm.013}$	163%

Table 8: Four PG-19 books of 200k tokens, read book after book. Loss, gain over frozen Pythia-14M (± SE over 4 books, paired) and share of the gap between frozen 14M and frozen 31M closed. Dynamic-evaluation runs: plain SGD on all parameters.

evaluation (plain SGD, lr 3×10^{-4}) gains about as much ($+0.168\pm0.013$ over its frozen loss), stays ahead of 14M with dynamic evaluation ($+0.348\pm0.013$), and does not reach a frozen Pythia-70M ($+0.054\pm0.010$ behind).

5 Discussion

Does dynamic evaluation help more with locality than with random input? Yes: the gain from more than 1k tokens back mostly disappears when the chunks are shuffled, and the overall gain falls at every step from long books to shuffled chunks (Sections 4.1 and 4.3).

Does it evict like a cache? Yes for the memory of a specific book: it halves after a few thousand tokens of other books, with a long tail (Section 4.4). Unlike a cache, filling it overwrites general knowledge outside the domain: WikiText forgetting is about twice the gain on the stream, and within 14M and 31M a stronger cache forgets more (Sections 4.5 and 4.7).

Is the effect larger in small models? Not between 14M and 31M: locality does not change beyond noise, and 14M does not pay more forgetting per nat of gain. 70M has less cache, but how much depends on the update rule and the step size it allows (Section 4.6).

Is there a knob? Partly: the parameter subset and the update rule move the size of the cache, but within 14M and 31M every setting we tried moves cache and forgetting together (Section 4.7). No knob tested here separates the cache from general knowledge.

Can a small model with dynamic evaluation replace a larger one? Not at these sizes: on long books dynamic evaluation is worth about half the step from 14M to 31M, but 31M gains as much from it (Section 4.8).

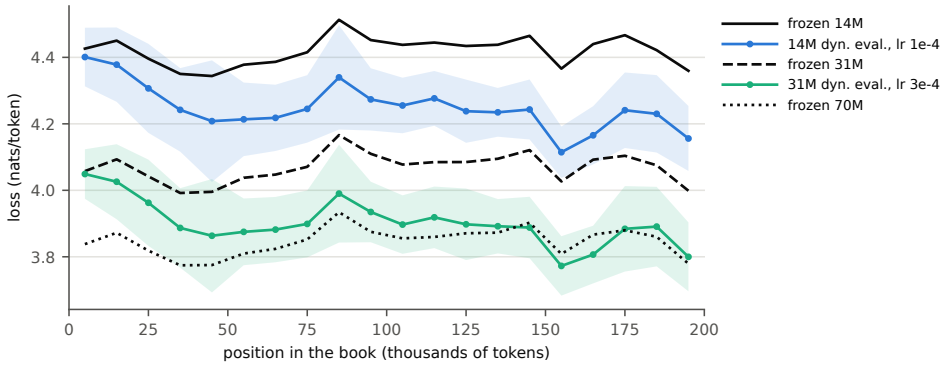


Figure 6: Loss by position in 200k-token books (10k-token bins, mean of 4 books). Bands: ±1 SE over books, which mostly reflect differences in book difficulty; the paired differences in Table 8 are much tighter.

Baselines with the full context window and with matched compute are still missing.

Does it resemble human memory? Only suggestively, and we did not test it directly. Most of a book's memory is on the specific tokens seen (+0.223 for rare tokens of A1, against +0.056 for other rare tokens), which resembles episodic memory, while the gain that survives shuffling transfers to unseen books, which resembles semantic memory. Eviction is fast at first and slow later, closer to a power law than to an exponential.

6 Limitations

- **One seed.** Error bars are across books, not across models or runs.
- **Three sizes of one family.** Pythia-14M, 31M and 70M only.
- **One kind of data.** PG-19 books only; WikiText is the only out-of-domain text, and forgetting is measured on 76 WikiText chunks and 6 books.
- **No full-context baseline.** Chunks of 256 tokens are scored without earlier context; a model with its full context window would see more of the recent past through attention.
- **Learning rates.** Rates are picked on mean validation loss, not on locality, and different subsets and sizes use different rates, so comparisons across them mix in the rate.
- **Wall time is not compute.** We did not measure compute; dynamic-evaluation runs update after every chunk, frozen runs do not.
- **Overlapping sets.** All 5 long books and 2 of the 4 very long books are also in the main set.

7 Future work

- A separate fast cache part on top of a frozen or slowly learning model, with a fallback to the frozen model when the cache does not help.
- Capacity: many books interleaved at once, and the size of the adapted part against the eviction half-life.
- Why Adam removes the cache: ablate its momentum and its per-weight normalization.
- Baselines with the full context window and with matched compute.
- More seeds (PolyPythias [40]) and larger models.
- Direct memory tests: savings on re-reading, interference between similar books, spacing, and planted facts.

8 Conclusion

Small language models are cheaper to run and easier to deploy, but their limited capacity restricts how much knowledge and context they can retain. We started from the observation that what a model reads has temporal locality, so that a small model whose weights cached its current context, instead of permanently holding everything it might need, could perform on that context closer to a larger one. Dynamic evaluation is a natural candidate for such a cache, provided that its forgetting, usually seen as the price of learning at inference time, acts as eviction: a model that forgets old material as it learns new material might hold the recent context in its weights beyond the attention window. We asked whether this behaviour arises on its own in a small transformer that takes plain gradient steps as it reads, and tested the three properties a cache should have.

The first two hold. The benefit of dynamic evaluation depends on locality: most of the gain on tokens last seen in an earlier chunk requires the text to be read in order (only 20% of the gain less than 1k tokens back survives shuffling), the memory is tied to the book being read, and the gain shrinks steadily as books are interleaved. Other text evicts this memory, halving it after about 6.1k tokens of other books, with a long tail. The memory sits mostly in the MLPs.

The third property holds only in part. Knowledge of the same domain is largely preserved, but the cache shares its weights with general knowledge, and filling it costs knowledge outside the domain: about 2.0 nats of WikiText loss per nat gained. The update rule and the set of weights being updated change how much cache a model gets, but within 14M and 31M every setting we tried moved cache and forgetting together.

A standard transformer thus has a cache in its weights, but not a clean one: the useful forgetting that evicts old context comes bundled with the harmful forgetting of general knowledge. Even so, the cache already matters at this scale: on long books, a 14M-parameter model with dynamic evaluation closes 49% of the gap to a frozen model about twice its size, and the cache behaves in the same way at both sizes. Separating useful from harmful forgetting, for instance with a dedicated fast part of the network, could strengthen the cache and make better use of temporal locality, moving towards small models that perform on the context at hand like larger ones.

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